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UNIVERSITÄT
MAGDEBURG



FAKULTÄT FÜR
WIRTSCHAFTSWISSENSCHAFT

Otto-von-Guericke University Magdeburg

Faculty of Economics and Management

Chair in E-Business

Seminar Paper

Summer term 2026

Economic Policy Uncertainty, Public Attention, and DAX Volatility

(A Germany-focused analysis using EPU, Google Trends, and DAX volatility)

GitHub Repository: <https://github.com/Shahriare23/newanalysis>

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Date of submission: 10.07.2026

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1. Introduction

Economic uncertainty matters for financial markets because it can change how investors, firms, and the public think about future economic conditions. When uncertainty rises, market participants may become more cautious, search for more information, and react more strongly to economic news (Baker S. R., 2016). This is especially relevant during periods of political and economic stress, when topics such as inflation, recession, unemployment, and financial stability become more visible in public discussion.

In recent years, researchers have increasingly used alternative data sources to study public attention and information flows in financial markets. One example is Google Trends, which shows how often people search for specific economic or financial terms over time (Liu, 2015). Google Trends does not directly measure investor beliefs or expectations, but it can be useful as a proxy for information seeking by the public. A higher frequency of searches for economic topics may therefore reflect higher attention to economic uncertainty.

This project explores the relationship between economic policy uncertainty, Google Trends search activity, and DAX volatility. Economic policy uncertainty is represented by the German news-based index. Google Trends is used to measure public search activity and attention. DAX volatility is used as a market-based indicator of instability in the German stock market. The general idea is that greater economic policy uncertainty may be associated with stronger public search activity and higher DAX volatility.

The project does not try to prove a direct causal relationship. Instead, it examines whether these variables move together and how they interact over time. Higher uncertainty may lead people to search more for economic information, but stronger market volatility may also raise public concern and increase search activity. For this reason, the project focuses on the association between uncertainty, public attention, and market volatility in the German market context.

2. Literature Review

Uncertainty, public attention, and stock market volatility are closely related to the literature on media and finance. If people are faced with higher economic uncertainty, they tend to observe and

analyse the situation in order to understand what is going on. They may read news, follow economic discussions, or search for terms such as inflation, recession, unemployment, or financial markets on Google. This is why Google Trends is used in this project as a proxy for public information-seeking behavior.

A large literature on media and finance makes this strategy sensible. (Tetlock, 2015) shows that media is not only a passive conduit of information. It can influence where investors pay attention, how they form beliefs, and how markets react to economic events. This is useful for this project because Google searches can be understood as part of the same information environment. They do not directly reveal what people believe, but they help show when economic issues become more salient for the public.

(Tetlock, 2015) also points out that finance research increasingly uses information-based datasets such as newspaper archives, internet search data, and social media posts. This matters here because the project does not observe public concern directly. Instead, it uses search intensity as a proxy. In short, when search interest in economic topics rises, it can be interpreted as people paying more attention to economic uncertainty. This does not mean that every search is made by an investor, but it is still a useful signal of public attention.

A second important strand of the literature is the measurement of economic policy uncertainty. (Baker S. R., 2016) construct the Economic Policy Uncertainty index using the frequency of newspaper articles that mention terms related to the economy, uncertainty, and policy. Their study is important because uncertainty cannot be observed directly in the same way as a stock price or an interest rate. It has to be measured indirectly. Their approach shows that text-based information can be used to quantify uncertainty.

(Liu, 2015) show that the EPU index rises during major political, financial, and economic events such as elections, wars, financial crises, and fiscal policy disputes. Such events usually increase uncertainty about future policy and economic developments. The authors also find that higher policy uncertainty is associated with greater stock price volatility and weaker investment and employment outcomes. This is relevant here because DAX volatility is used as a proxy for instability in the German stock market.

The logic of (Bock, 2018) is similar to the logic of this project. Their EPU index is based on newspaper coverage, while this project uses Google Trends to capture public search behavior. Both approaches rely on indirect indicators to measure concepts that cannot be observed perfectly. Google Trends does not measure fear, expectations, or beliefs in an ideal way, but it can indicate when people search more actively for economic information.

The behavioral side of the project is supported by (Barber, 2007). They argue that individual investors face a search problem when deciding which stocks to buy. There are too many candidate stocks in the market for investors to evaluate all of them carefully. As a result, investors are more likely to consider stocks that first attract their attention. These may be stocks in the news, stocks with unusually high trading volume, or stocks with very high or very low recent returns.

(Barber, 2007) find that individual investors are more likely to buy attention-grabbing stocks. This is useful because it shows that attention can matter for financial-market outcomes. Investors do not only respond to fundamental information; they also respond to information that is visible or salient. Although their paper focuses on stock purchases, the wider idea is relevant here. If people search more for economic topics during uncertain periods, this may reflect a broader increase in public attention.

Google Trends also has to be interpreted carefully. (Challet, 2014) show that attention matters more for individual, non-professional investors than for institutional investors. Professional investors usually have more information, better tools, and more formal investment processes. For this reason, Google Trends should not be treated as a measure of institutional analysis. It is more reasonable to treat it as a broad public or retail-level attention measure.

Taken together, the six papers support the mechanism of this project. (Baker S. R., 2016) provide the uncertainty side. (Tetlock, 2015) provides the media and information-flow perspective. (Barber, 2007) provide the behavioral explanation for why attention may matter. Building on these ideas, this project studies whether elevated economic policy uncertainty is associated with higher public search activity and higher DAX volatility.

3. Project Mechanism and Hypotheses

The mechanism of this project is based on the relationship between economic policy uncertainty, public information search, and stock market volatility. The basic idea is simple. When economic policy uncertainty increases, people may become more concerned about future economic conditions. In response, they may try to collect more information. This can appear as higher Google search activity for economic and financial terms.

The project uses Google Trends as a noisy but useful measure of public attention. The claim is not that every Google search is conducted by an investor, or that search activity measures investor expectations one-to-one. Instead, Search Index is treated as a broad signal of public attention and information demand. If search intensity rises during uncertain periods, this suggests that uncertainty has become more visible to the public.

Financial market volatility may also be part of this mechanism. When investors face an unclear economic-policy environment, they may react more strongly to news and adjust expectations more often. This can lead to larger price movements in the stock market. In this project, DAX Volatility captures these fluctuations in the German stock market.

Since we want to observe the short-run adjustments and long-run equilibrium relationships between the variables, we use a Vector Error Correction Model (VECM) replacing the initial VAR framework. This is useful because it allows the variables to influence each other over time instead of forcing the analysis into a simple one-way model. The project is guided by three expectations.

Hypotheses Expected Relationship

H_1 : Increases in media intensity (Germany News Index (NI_t)) lead to subsequent increases in Information Seeking behavior (SI_t).

H_2 : Financial market volatility Granger causes/ increases information-seeking behavior.

H_3 : Information-seeking behavior predicts or Granger causes financial market volatility (V_t).

These expectations are not interpreted as proof of causality. They are used to organize the empirical analysis and to check whether the variables contain useful information about each other over time.

4. Data, Variables and Measurement

The analysis uses data for Germany from January 2010 to January 2026. We chose this time period because it is the range that remained after matching data sources at the same monthly frequency. Some data series, such as the EPU index, are available for a longer period, but we could not match all variables for the whole range. The dataset has 193 observations and includes these main variables:

- Month
- Germany_News_Index = NI_t
- Search_Index = SI_t
- DAX_Volatility = V_t

Inflation, unemployment, and ECB-related terms are included in the Google Trends keyword set as components of public economic attention. They are not treated as separate macroeconomic control variables in the final dataset. We do not fill in missing observations. This means the sample gets shorter. The values in the dataset are based on observed data. We prefer a monthly sample over a longer dataset with artificial values. The main uncertainty variable is the news-based Economic Policy Uncertainty index, which we call Germany News Index. We use it as a measure of economic policy uncertainty in Germany based on newspaper coverage.

The Search Index is made from Google Trends data for Germany. We use Google search activity as a proxy for attention, following (DA, 2011), who propose Google Search Volume Index as a direct measure of investor attention. The keyword choice is also guided by (Ayala, 2024) whose systematic review shows that GSVI-based stock-market studies select keywords according to the search term, market region, and data frequency.

The Search Index is calculated as the simple arithmetic average of thirteen Google Trends keyword series: inflation, recession, unemployment, interest rates, ECB, ECB interest rate, monetary policy ECB, Germany GDP, market crash, banking crisis, DAX, stock market, and stock prices –

$$SI_t = \frac{1}{K} \sum_{k=1}^K G T_{k,t}, \quad K = 13$$

We categorize the 13 keywords to inspect macroeconomic uncertainty, monetary-policy attention, financial-stress attention, and financial-market attention in the following table –

Category	Keywords Included	Conceptual Dimension Captured	Rationale for Inclusion
Policy-Related Attention	1. ECB 2. ECB interest rate 3. Monetary policy ECB 4. Interest rates	Attention toward monetary policy decisions and central bank.	These keywords capture public interest in monetary policy developments, particularly European Central Bank decisions and interest rate changes that influence economic uncertainty and expectations.
Financial and Market Stress	5. DAX 6. Stock Market 7. Stock Prices 8. Market crash 9. Banking crisis	Attention toward financial instability and market risk.	These keywords represent concerns related to financial market disruptions, banking sector uncertainty, and potential economic risks.

General Macroeconomic Conditions	10. Inflation	Attention toward	These keywords capture broader
	11. Recession	macroeconomic	concerns regarding inflationary
	12. Unemployment	performance and	pressure, economic downturns,
	-nt	economic	labor market conditions, and
	13. Germany	conditions.	national economic performance.
	GDP		

Table – 1 Classification of Google Trends Keywords Used to Construct the Information-Seeking Index

Google Trends values are normalized from 0 to 100. Therefore, the Search Index shows whether search interest was relatively high or low over time, not the absolute number of searches. We use the index as a proxy for attention and information-seeking behavior.

DAX volatility is the financial-market variable. We use daily DAX closing prices, calculate daily log returns, and then calculate the standard deviation of daily returns within each month-

$$r_d = \ln\left(\frac{P_d}{P_{d-1}}\right)$$

$$V_m = \sqrt{\frac{1}{D_m - 1} \sum_{d=1}^{D_m} (r_{d,m} - \bar{r}_m)^2}$$

This gives a monthly measure of stock-market volatility. The dataset connects the three parts of the research question: the EPU index measures the uncertainty environment, the Search Index measures information demand, and DAX volatility measures stock-market fluctuations.

This makes the dataset suitable for examining whether months with higher economic policy uncertainty are also associated with stronger search activity and higher DAX volatility.

Variable Construction

The first main variable is the Economic Policy Uncertainty index for Germany. In the dataset, this variable is called Germany News Index. It is used at a monthly frequency and measures economic policy uncertainty based on German newspaper coverage.

The second main variable is the Google Trends Search Index. The Search Index is calculated as the simple arithmetic average of thirteen Google Trends keyword series for Germany: inflation, recession, unemployment, interest rates, ECB, ECB interest rate, monetary policy ECB, Germany GDP, market crash, banking crisis, DAX, stock market, and stock prices. These keywords cover macroeconomic uncertainty, monetary-policy attention, financial-stress attention, and financial-market attention.

The third main variable is DAX volatility. Daily DAX closing prices are used to calculate daily log returns. Monthly DAX volatility is then calculated as the standard deviation of daily log returns within each month.

The final dataset contains Month, Germany News Index, Search Index and DAX Volatility.

5. Empirical Strategy

The empirical analysis uses three monthly indicators from the period 2010 to 2026 – the Germany News Attention Index, Google Trends-based Search Index, and DAX volatility. We treat news attention, search behavior, and market volatility as interconnected time-series processes that may influence each other over time. The News Attention Index captures the intensity of public and media focus on economic developments. The Search Index represents online information-seeking behavior. DAX volatility is used as a measure of financial market uncertainty.

We constructed a time-series econometric framework for this analysis. At first, we conducted the Augmented Dickey–Fuller (ADF) tests to examine the stationarity properties of each variable. Since non-stationary variables may produce misleading regression relationships, the variables are tested at levels and after first differencing. The results (Table- 5 and 6, Appendix) indicate that Germany News Index and Search Index are integrated of order one, while DAX volatility exhibits stationary behaviour at level.

Given the presence of long-run relationships among the variables, we employ the Johansen trace test (Table- 7, Appendix) to determine the number of cointegrating relationships. The test identifies two cointegrating vectors justifying the use of a Vector Error Correction Model (VECM) in lieu of the Vector Autoregression (VAR) Model. The VECM framework is suitable because it captures both short-run adjustments and long-run equilibrium relationships between the variables.

Our estimated VECM includes three lagged differences and two cointegrating relationships:

$$\Delta Y_t = \alpha \beta' Y_{t-1} + \sum_{i=1}^3 \Gamma_i \Delta Y_{t-1} + \varepsilon_t$$

where,

Y_t = The vector containing News Attention, Search Behaviour, and DAX Volatility. The vector

β = The matrix containing the cointegrating vectors. It captures long-run equilibrium relationships.

α = The coefficient matrix that measures the speed of adjustment toward equilibrium after a temporary shock.

Γ_i = The matrices that captures short-run dynamics at lag order.

To further examine the dynamic transmission mechanism, impulse response functions (IRFs) are estimated over a 12-month horizon. IRFs identify how each variable responds to shocks originating from the other variables. Finally, Granger causality tests are conducted within the VECM framework to determine whether past values of one variable provide statistically significant predictive information about another. Together these methods allow the study to distinguish between short-run dynamics, long-run equilibrium adjustments and directional predictive relationships.

6. Findings and Analysis

This section presents the empirical findings in three stages. First, time-series properties and long-run relationships among the variables are examined through stationarity and cointegration tests. Then we construct the Vector Error Correction Model (VECM) to identify short-run dynamics and

long-run adjustment mechanisms. Finally, we evaluate the hypotheses using VECM coefficients, impulse response functions (IRFs), and Granger causality tests.

6.1 Stationarity and Long-Run Relationship

The Augmented Dickey–Fuller (ADF) (Table -5) tests indicate that Germany News Index and Search Index are non-stationary at levels but become stationary after first differencing, confirming that both variables follow an I(1) process. DAX volatility is stationary at level, reflecting the mean-reverting characteristics commonly observed in financial volatility measures.

The Johansen trace test (Table-7) confirms the presence of long-run equilibrium relationships among the variables. The null hypothesis of no cointegration is rejected, with the trace statistic exceeding the critical value. The estimated cointegration rank of two indicates that media attention, online search behaviour, and market volatility share long-run interactions. Therefore, a Vector Error Correction Model (VECM) with three lagged differences and two cointegrating relationships is estimated.

6.2 VECM Estimation and Dynamic Adjustment

The VECM results indicate that the interaction among the three variables is primarily driven by an attention transmission process rather than a market-impact process.

In the Germany News Index equation, lagged DAX volatility has a positive and statistically significant effect (coefficient = 5184.87, $p = 0.008$). This suggests that periods of increased market uncertainty generate greater media attention. Therefore, financial market instability appears to act as a trigger for public and media awareness.

The Search Index equation provides stronger evidence of information transmission. Lagged Germany News Index significantly influences search behaviour (coefficient = -0.4184, $p < 0.001$), indicating that changes in media attention affect subsequent online information-seeking activity. Search behaviour also demonstrates persistence, as its own lagged values remain statistically significant. Additionally, the third lag of DAX volatility significantly affects Search Index (coefficient = 209.14, $p = 0.011$), suggesting that market uncertainty influences information-seeking behaviour with a delayed response.

Det. terms outside the coint. relation & lagged endog. parameters for equation Germany_News_Index

	coef	std err	z	P> z	[0.025	0.975]
L1.Germany_News_Index	-0.4145	0.088	-4.706	0.000	-0.587	-0.242
L1.Search_Index_v3	-3.6310	2.145	-1.693	0.091	-7.835	0.573
L1.DAX_Volatility	5184.0867	1942.741	2.668	0.008	1376.383	8991.790
L2.Germany_News_Index	-0.3654	0.087	-4.219	0.000	-0.535	-0.196
L2.Search_Index_v3	1.0843	2.193	0.494	0.621	-3.213	5.382
L2.DAX_Volatility	3661.8129	1900.901	1.926	0.054	-63.885	7387.511
L3.Germany_News_Index	-0.0735	0.081	-0.911	0.362	-0.232	0.085
L3.Search_Index_v3	-0.8590	2.626	-0.327	0.744	-6.005	4.287
L3.DAX_Volatility	2370.2494	1804.467	1.314	0.189	-1166.442	5906.941

Det. terms outside the coint. relation & lagged endog. parameters for equation Search_Index_v3

	coef	std err	z	P> z	[0.025	0.975]
L1.Germany_News_Index	-0.0169	0.004	-4.239	0.000	-0.025	-0.009
L1.Search_Index_v3	0.4043	0.097	4.159	0.000	0.214	0.595
L1.DAX_Volatility	-4.1384	88.041	-0.047	0.963	-176.695	168.418
L2.Germany_News_Index	-0.0124	0.004	-3.163	0.002	-0.020	-0.005
L2.Search_Index_v3	0.1993	0.099	2.005	0.045	0.005	0.394
L2.DAX_Volatility	52.0494	86.144	0.604	0.546	-116.791	220.889
L3.Germany_News_Index	-0.0194	0.004	-5.303	0.000	-0.027	-0.012
L3.Search_Index_v3	-0.0413	0.119	-0.347	0.729	-0.274	0.192
L3.DAX_Volatility	209.1448	81.774	2.558	0.011	48.870	369.419

Det. terms outside the coint. relation & lagged endog. parameters for equation DAX_Volatility

	coef	std err	z	P> z	[0.025	0.975]
L1.Germany_News_Index	-3.051e-06	4.07e-06	-0.750	0.453	-1.1e-05	4.92e-06
L1.Search_Index_v3	4.158e-06	9.91e-05	0.042	0.967	-0.000	0.000
L1.DAX_Volatility	-0.0535	0.090	-0.596	0.551	-0.229	0.122
L2.Germany_News_Index	-3.371e-06	4e-06	-0.843	0.399	-1.12e-05	4.47e-06
L2.Search_Index_v3	0.0001	0.000	1.350	0.177	-6.17e-05	0.000
L2.DAX_Volatility	0.0119	0.088	0.135	0.892	-0.160	0.184
L3.Germany_News_Index	1.955e-06	3.72e-06	0.525	0.600	-5.34e-06	9.26e-06
L3.Search_Index_v3	-6.433e-05	0.000	-0.531	0.596	-0.000	0.000
L3.DAX_Volatility	0.0823	0.083	0.988	0.323	-0.081	0.246

Table – 2.1 VECM Analysis outcome.

In contrast, the DAX volatility equation shows limited evidence that attention variables influence market movements. Neither Germany News Index nor Search Index produces statistically significant effects on subsequent volatility changes. This indicates that public attention responds to financial uncertainty rather than acting as a direct source of market volatility.

The error correction coefficients further clarify the adjustment mechanism. The adjustment coefficients for Germany News Index and DAX volatility are statistically insignificant, suggesting that these variables do not actively correct deviations from long-run equilibrium. However, the Search Index adjustment coefficients are significant for both cointegrating relationships ($p < 0.001$), indicating that online information-seeking behaviour is the primary variable adjusting toward long-run equilibrium after temporary shocks.

Loading coefficients (alpha) for equation Germany_News_Index						
	coef	std err	z	P> z	[0.025	0.975]
ec1	-0.0586	0.048	-1.233	0.218	-0.152	0.035
ec2	1.8881	1.846	1.023	0.306	-1.731	5.507
Loading coefficients (alpha) for equation Search_Index_v3						
	coef	std err	z	P> z	[0.025	0.975]
ec1	0.0108	0.002	5.033	0.000	0.007	0.015
ec2	-0.4374	0.084	-5.228	0.000	-0.601	-0.273
Loading coefficients (alpha) for equation DAX_Volatility						
	coef	std err	z	P> z	[0.025	0.975]
ec1	-5.854e-08	2.19e-06	-0.027	0.979	-4.36e-06	4.24e-06
ec2	-8.839e-05	8.53e-05	-1.037	0.300	-0.000	7.87e-05
Cointegration relations for loading-coefficients-column 1						
	coef	std err	z	P> z	[0.025	0.975]
beta.1	1.0000	0	0	0.000	1.000	1.000
beta.2	-3.28e-15	0	0	0.000	-3.28e-15	-3.28e-15
beta.3	1.656e+05	3.21e+04	5.164	0.000	1.03e+05	2.28e+05
const	-2070.5423	831.280	-2.491	0.013	-3699.822	-441.263
Cointegration relations for loading-coefficients-column 2						
	coef	std err	z	P> z	[0.025	0.975]
beta.1	4.022e-18	0	0	0.000	4.02e-18	4.02e-18
beta.2	1.0000	0	0	0.000	1.000	1.000
beta.3	4282.4451	377.611	11.341	0.000	3542.342	5022.548
const	-59.1841	9.790	-6.046	0.000	-78.371	-39.997

Table 2.2: VECM Analysis outcome (co-efficients).

Furthermore, the IRF analysis ensured that our VECM is stable and stationary in behavior not explosive. We demonstrate the IRF tests result in the following-

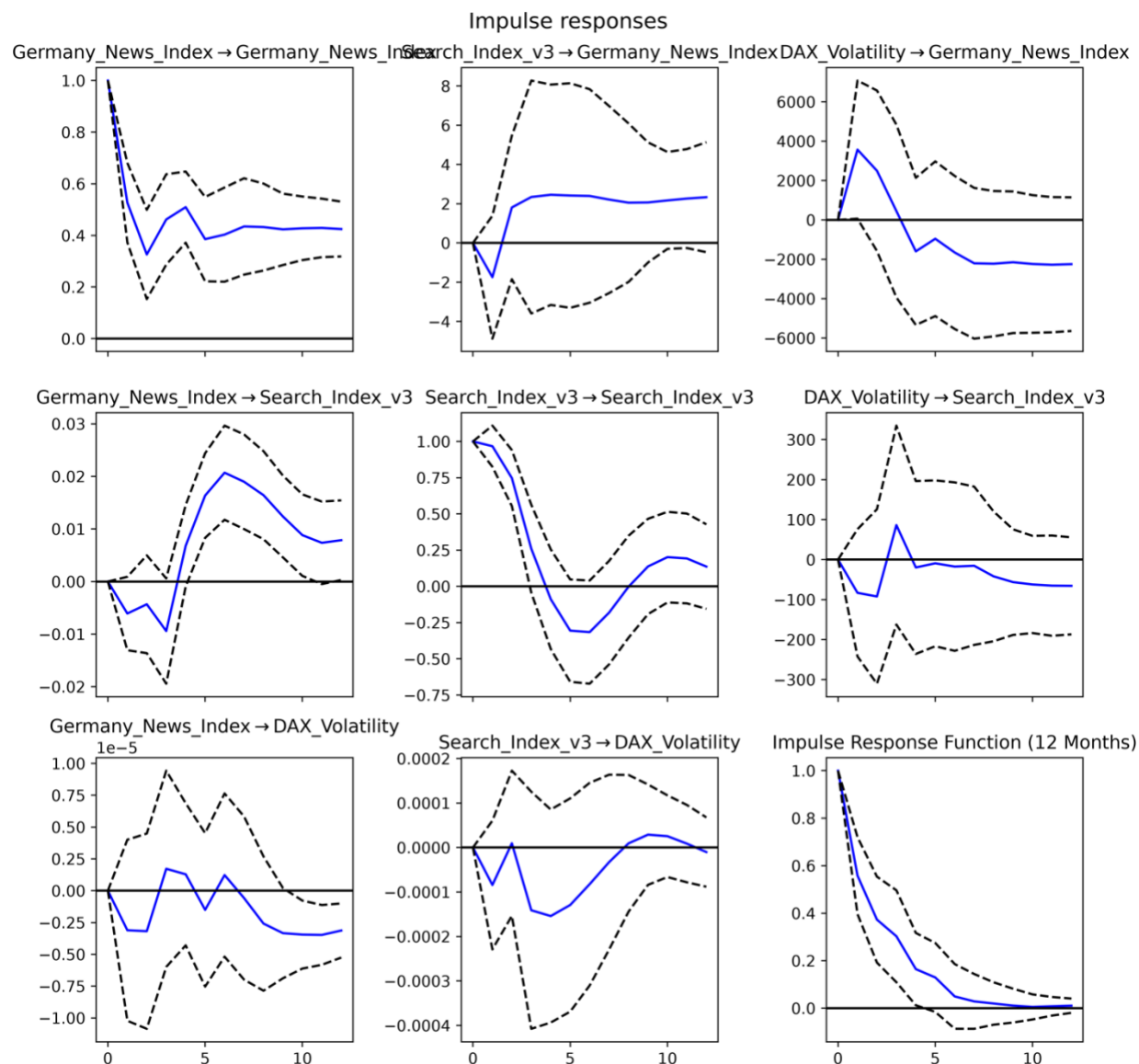


Figure – 1: Impulse Response Functions.

6.3 Hypothesis Testing and Empirical Evidence

H₁: Media intensity increases information-seeking behavior.

The results provide strong support for H₁. The VECM estimation demonstrates that changes in Germany News Index significantly influence subsequent movements in Search Index. The Granger causality analysis confirms a directional predictive relationship from media attention to online search activity:

Germany News Index



Search Index / Information Seeking

The relationship is statistically significant ($F = 7.6443$, $p < 0.001$), indicating that previous changes in media attention contain predictive information about future search behaviour.

The impulse response analysis provides additional support. A positive shock to Germany News Index generates a positive response in Search Index after an initial adjustment period, suggesting that increased media exposure encourages individuals to seek additional information.

Therefore, media attention acts as an important driver of public information-seeking behaviour.

H₂: Financial market volatility increases information-seeking behavior.

The findings provide partial support for H₂. The VECM results indicate that market volatility influences search behaviour with a delayed effect. Specifically, the third lag of DAX volatility has a positive and statistically significant effect on Search Index (coefficient = 209.14, $p = 0.011$). This suggests that market instability does not immediately translate into increased information demand but emerges after a short adjustment period.

The Granger causality test provides weaker evidence. The causal direction from DAX volatility to Search Index is marginally significant ($F = 2.1424$, $p = 0.0744$), indicating that the predictive relationship exists only at the 10% significance level.

The impulse response functions similarly show that volatility shocks influence search behaviour, although the magnitude of the response is smaller compared with the effect of news attention.

Overall, the evidence suggests that financial uncertainty contributes to public information-seeking behaviour, but the effect is weaker and more delayed compared with the influence of media attention.

Decision: H₂ is partially supported.

H₃: Information-seeking behavior predicts financial market volatility.

The results do not support H₃. The VECM estimation provides no significant evidence that changes in Search Index influence subsequent DAX volatility. Similarly, the Granger causality test fails to identify a predictive relationship:



The relationship is statistically insignificant ($F = 0.7395$, $p = 0.5653$). The impulse response analysis also shows limited and unstable responses of DAX volatility following shocks to Search Index.

These findings indicate that online information-seeking behaviour reflects reactions to uncertainty rather than an independent mechanism driving financial market instability. Although individuals may increase their search activity during uncertain periods, such behaviour does not translate into measurable changes in DAX volatility.

Decision: H₃ is not accepted.

6.4 Summary of Hypothesis Testing Results

Hypothesis	Expected Relationship	Empirical Evidence	Statistical Evidence	Decision
H ₁	Germany News Index → Search Index	Media attention increases subsequent online information seeking behaviour	Granger causality: $F = 7.6443$, $p < 0.001$; VECM and IRF support relationship	Accepted

H₂	DAX Volatility → Search Index	Market volatility affects search behaviour with delayed effects.	VECM: L3.DAX volatility coefficient = 209.14, p = 0.011; Granger p = 0.0744	Partially accepted.
H₃	Search Index → DAX Volatility	Search activity does not significantly influence market volatility.	Granger causality: F = 0.7395, p = 0.5653; VECM and IRF are insignificant.	Rejected

Table 3 – Summary of the findings at a glance.

6.5 Overall Interpretation

The empirical evidence indicates that the relationship between uncertainty, attention, and financial markets follows an attention-generation mechanism rather than an attention-driven market mechanism. The strongest relationship identified in the model is the transmission from media attention to information-seeking behaviour. Market volatility also contributes to attention formation, particularly through delayed effects, but online search activity does not significantly influence subsequent market volatility.

The findings suggest the following dynamic pathway:



rather than:



Thus, digital attention indicators appear to function primarily as measures of public response to economic uncertainty rather than as independent predictors of financial market movements.

We can summarize the IRF tests result in the following- In summary, the IRF analysis ensured that our VAR is stable and stationary in behavior not explosive.

7. Limitations

The results should be interpreted with caution. The analysis is based on monthly data, so it may miss short-lived reactions within days or weeks. Google Trends is only a proxy for attention and does not show who searched, why they searched, or whether searches reflected investment decisions. The Search Index also depends on the selected keywords. In addition, Germany News Index captures uncertainty through newspaper coverage, which may react to events rather than only measure them. Finally, the sample includes major shocks such as COVID-19, the Russia-Ukraine war, the inflation surge, and ECB tightening, which may make the relationship between uncertainty, attention, and DAX volatility less stable over time.

8. Conclusion

This project examines the relationship between economic policy uncertainty, public information-seeking behavior, and DAX volatility in Germany. The analysis combines a news-based uncertainty index, a Google Trends Search Index, and DAX volatility at a monthly frequency from January 2010 to January 2026. The results provide partial support for the attention channel. The strongest finding is that Germany News Index predicts Search Index. This suggests that when uncertainty becomes more visible in the news environment, the public searches more actively for economic and financial information. At the same time, the analysis does not find strong evidence that Search Index predicts DAX volatility. Overall, the project shows that media-based uncertainty and public search behavior are connected. However, the connection between public attention and market volatility is weaker than expected. This makes the results more realistic. They suggest that

Google Trends is useful for studying public attention, but it should not be treated as a direct driver of financial-market volatility.

9. AI Assistance Statement

This seminar paper involved limited and disclosed AI assistance. AI tools were used to support language editing, improve section structure, help formulate table and figure templates, and generate draft wording for methodological explanations and code comments. AI tools were also used as a programming assistant for cleaning and organizing Python scripts used to construct the Search Index, merge datasets, and produce exploratory figures. All research decisions, including the choice of topic, conceptual framing, variable selection, source selection, econometric design, interpretation of results, and final written conclusions, were made and verified by the group. All numerical results, regressions, figures, and dataset constructions were checked against the underlying data files and scripts before inclusion in the paper. No AI-generated citation or empirical claim was included without manual verification.

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11. Appendix

ADF Test - Levels

Variable	ADF Statistic	p-value	Conclusion
Germany_News_Index	0.5376	0.98597	Non-stationary
Search_Index_v3	0.7594	0.99095	Non-stationary
DAX_Volatility	-5.7463	0.0	Stationary

Table – 5: ADF Test – Levels

ADF Test - First Difference

Variable	ADF Statistic	p-value	Conclusion
Germany_News_Index	-5.1032	1e-05	Stationary
Search_Index_v3	-3.6132	0.00551	Stationary
DAX_Volatility	-5.3163	1e-05	Stationary

Table 6: ADF Test – First Difference.

Trace Statistics	90% Critical Value	95% Critical Value	99% Critical Value
59.3316	27.0669	29.7961	35.4628
27.0042	13.4294	15.4943	19.9349
0.2111	2.7055	3.8415	6.6349

Table – 7: Johansen Cointegration Test results.

Causal Direction	Test Statistic	p-value	Conclusion
Germany News Index → DAX Volatility	0.6514	0.62613	Not Significant
Search Index → DAX Volatility	0.7395	0.56532	Not Significant
DAX Volatility → Germany News Index	2.3603	0.05240	Not Significant
DAX Volatility → Search Index	2.1424	0.07441	Not Significant
Germany News Index → Search Index	7.6443	0.00001	Significant
Search Index → Germany News Index	1.7464	0.13844	Not Significant

Table – 8 : Granger Causality Test Results (VECM).